

CUSTOMER SUPPORT TICKET PRIORITY PREDICTION AND AUTOMATED ASSIGNMENT SYSTEM USING AGENTFORCE.

User Story:

The Customer Support Ticket Priority Prediction and Automated Assignment System uses Salesforce and Agentforce to analyze incoming customer support ticket descriptions, automatically determine their priority level as High, Medium, or Low, and execute automated workflows for rapid resolution. Built around the custom Support Ticket Intelligence object and an Auto-Launched Flow, the system evaluates key urgency markers, identifies SLA breach risks, and assigns tickets directly to the appropriate support agents. High-priority tickets trigger immediate task creation to ensure prompt action, while role-based security and tailored permissions provide managers with real-time operational oversight and performance tracking across the support lifecycle.

1. INTRODUCTION

1.1 Project Overview:

The **Intelligent Ticket Prioritization and Routing System** is a Salesforce- and Agentforce-driven solution designed to modernize high-volume customer service operations. By transitioning from manual triage to autonomous classification and assignment, the platform evaluates incoming customer requests, dynamically determines urgency (High, Medium, Low), routes tickets to the appropriate support agents, and triggers follow-up actions for critical issues. Built using a custom Salesforce data model, automated Flow engines, and Agentforce topics, the system bridges the gap between customer intake and rapid resolution..

1.2 Purpose:

- **Accelerate First-Response Times:** Eliminate administrative intake bottlenecks by classifying and routing incoming issues instantly.
- **Guarantee Critical-Issue Prioritization:** Identify high-risk issues and SLA breach threats through automated keyword and semantic evaluation, generating urgent tasks for critical tickets.
- **Balance Agent Workloads:** Replace ad-hoc manual assignment with consistent, rules-based routing to maximize support team productivity.
- **Enhance Operational Visibility:** Provide support managers with real-time tracking across ticket lifecycles, team allocation, and SLA compliance via centralized dashboards.

2. IDEATION PHASE

PS ID	I am (User Persona)	I'm trying to	But	Because	Which makes me feel
PS-1	Support Agent	Focus on resolving critical	Critical tickets wait in a generic queue without	Ticket prioritization is done	Frustrated and reactive

		customer outages first	instant priority tags	manually line-by-line	
PS-2	Support Manager	Ensure SLA compliance on high-impact customer issues	Urgent issues get delayed behind older low-priority requests	High ticket volume creates assessment delays	Concerned about customer escalations
PS-3	Support Manager	Assign technical failure tickets directly to senior staff	Reassignments between general and senior queues cause delays	Initial routing lacks automated intelligence	Overwhelmed by administrative overhead
PS-4	Support Agent	Catch urgency signals in customer descriptions	Key phrases like "failure" or "not working" get missed during busy shifts	Descriptions are unstructured text of varying lengths	Uncertain and stressed
PS-5	Customer (kows)	Get rapid intervention for a broken system	Response times lag behind the actual business impact of the outage	Manual triage stands between ticket submission	Dissatisfied and ignored

2.1

Problem Statement (Customer Problem Statement Template)

2.2 Empathy Map Canvas

User Persona: Support Agent

- **Think & Feel:** Wants critical issues identified quickly so urgent customer problems receive attention.
- **Hear:** Receives customer issue descriptions and support requests.
- **See:** Ticket status, issue type, description, priority level, assignment, and SLA risk.
- **Say & Do:** Reviews tickets, handles assigned cases, and focuses on urgent support issues.
- **Pain:** Manual prioritization, missed urgency signals, and delays in handling critical tickets.

- **Gain:** Automatic priority classification, clearer assignment, urgent-task creation, and reduced manual effort.

2.3 Brainstorming & Idea Prioritization

- **Ideas Considered:** Manual ticket registers, Spreadsheet-based ticket tracking, Standalone support portal, Salesforce CRM with Agentforce.
- **Final Selection:** Salesforce CRM selected due to automation through Salesforce Flow, AI-assisted ticket analysis through Agentforce, automatic priority classification, automatic urgent-task creation, and reduced manual support effort.

PHASE -1

3. REQUIREMENT ANALYSIS PHASE

3.1 Customer Journey Map

Stages:

Ticket Creation → Account Identification (kows) → Ticket Retrieval (TKT-0001) → Description Analysis → Priority Classification (High) → Agent Assignment (Senior Support Agent) → Urgent Task Creation → Support Handling → Resolution.

3.2 Solution Requirements

Functional Requirements:

- **FR-1:** Support ticket creation and management.
- **FR-2:** Account and Contact association.
- **FR-3:** Ticket description analysis.
- **FR-4:** Automatic priority classification.
- **FR-5:** High-priority task creation.
- **FR-6:** Support agent assignment.
- **FR-7:** SLA breach risk checking.
- **FR-8:** Agentforce conversational analysis.

Non-Functional Requirements:

- **Usability:** Clear Agentforce interaction and understandable priority messages.
- **Reliability:** Consistent priority classification based on configured conditions.
- **Performance:** Quick retrieval and analysis of the latest support ticket.
- **Security:** Controlled Salesforce access to customer and ticket information.
- **Maintainability:** Flow and Agentforce configuration can be updated.
- **Scalability:** Supports increasing support-ticket volumes through Salesforce automation.

3.3 Data Flow Diagram

- **Flow:** Customer Request → Account (kows) → Latest Support Ticket (TKT-0001) → Ticket Description (urgent not working failure) → Analyze Description → Priority Level (High) → Agent Assignment (Senior Support Agent) → High-Priority Task Creation → Final Response.
- **Agentforce:** User provides Account Name → Agentforce Support Ticket Priority Analysis → Auto-Launched Flow → Ticket Analysis → Priority and Assignment Output → Response to User.

3.4 Technology Stack

- **Platform:** Salesforce
- **Database:** Support_Ticket_Intelligence__c with Account, Contact, User, and Task records
- **Logic:** Auto-Launched Flow (Support Ticket Intelligence - V2)
- **AI:** Agentforce
- **Agent Configuration:** Support Ticket Priority Analysis subagent
- **Automation:** Flow Decision, Assignment, and Create Records
- **Output:** Agentforce conversation response

4. PROJECT DESIGN PHASE

4.1 Problem-Solution Fit

Problem	Solution	Benefit
Manual ticket prioritization	Flow analyzes description and sets priority	Faster identification of urgent cases
Urgent issues may be missed	High-priority keyword decision	Critical cases receive attention
Manual task creation	Flow creates Task for High priority	Reduced manual effort
Manual support assignment	Flow sets assigned support level	More consistent handling
SLA risk can be overlooked	Optional SLA Risk decision	Better risk awareness
Need for conversational access	Agentforce subagent	AI-assisted interaction

4.2 Proposed Solution

- **Problem:** Manual ticket prioritization and assignment causing delays in handling critical customer support issues.
- **Solution:** Salesforce-based Customer Support Ticket Priority Prediction and Automated Assignment System.
- **Innovation:** Agentforce AI + Auto-Launched Flow for analyzing ticket descriptions, automatically classifying priority, assigning support level, and triggering urgent-task automation.
- **Impact:** Faster identification of critical tickets, reduced manual effort, improved response time, and better support-team productivity.
- **Scalability:** Supports increasing support-ticket volumes through Salesforce automation.

4.3 Solution Architecture

- **Components:** Users / Support Team → Agentforce Support Ticket Priority Analysis → Auto-Launched Flow → Account & Ticket Data → Description Analysis → Priority Decision → Task / Assignment → Final Message → Agentforce Response.

5. PROJECT PLANNING & SCHEDULING

5.1 Product Backlog and Sprint Schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority
Sprint-1	Developer Setup	USN-1	Configure the Salesforce developer environment for Support Ticket Intelligence.	3	High
Sprint-2	Data Modeling	USN-2	Create the Support Ticket Intelligence object with required fields.	5	High
Sprint-2	Data Modeling	USN-3	Associate support tickets with Accounts and Contacts (kows).	3	High
Sprint-3	Automation	USN-4	Auto-Launched Flow to retrieve latest ticket and analyze description.	5	High

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority
Sprint-3	Automation	USN-5	Automatically classify tickets as High, Medium, or Low based on keywords.	5	High
Sprint-4	Agentforce & Automation	USN-6	Agentforce analyzes ticket details and triggers backend Flow.	8	High
Sprint-4	Task & Assignment	USN-7	Create urgent task and assign to Senior Support Agent for High priority.	5	High
Sprint-5	SLA Management	USN-8	Check SLA breach risk for older unresolved tickets.	5	Medium
Sprint-6	Conversational Support	USN-9	Conversational interaction with Agentforce returning ticket parameters.	5	Medium

PHASE -2

6. PROJECT DEVELOPMENT PHASE

Milestone 1: Salesforce Setup & Developer Account

Developer edition organization configured and verified on the Salesforce Platform.

The image shows two screenshots related to setting up a Salesforce Developer Edition account.

Top Screenshot: Salesforce Developer Edition Sign-up Page

Build for the Agentic Enterprise. Sign up for your Developer Edition.

The entire Salesforce platform - exposed as APIs, MCP tools, and CLI commands. Build apps, ship agents, and let coding agents do the heavy lifting with Salesforce Headless 360.

- ✓ Build AI agents with Agentforce, grounded in your data with Data 360
- ✓ Connect Claude Code, Cursor, or Codex to your org with hosted MCP
- ✓ Build in the browser with the Agentforce Vibes IDE - no setup required
- ✓ Drag-and-drop when you want it, Apex when you need it
- ✓ 60+ MCP tools and 30+ coding skills, ready out of the box

Sign up for your Developer Edition

A free Salesforce Platform environment with Agentforce and Data 360

First name: Kowsalya ✓ Last name: R ✓

Job title: Company:

Country/region: India ✓ Work email:

☐ I agree to the [Main Services Agreement - Developer Services](#), [Salesforce Program Agreement](#), and [API Terms of Service](#)

Your free trial may be provisioned on or migrated to Hyperforce, Salesforce's public cloud infrastructure.

I acknowledge, as described in the Developer Documentation: (1) the Developer Edition includes autonomous and other generative AI features; and (2) Salesforce may limit use of those features and the org, and may terminate any org that has been inactive for 45 days.

We value your privacy. To learn more, visit our [Privacy Statement](#).

Sign me up

Bottom Screenshot: Gmail Email

Welcome to your Developer Edition

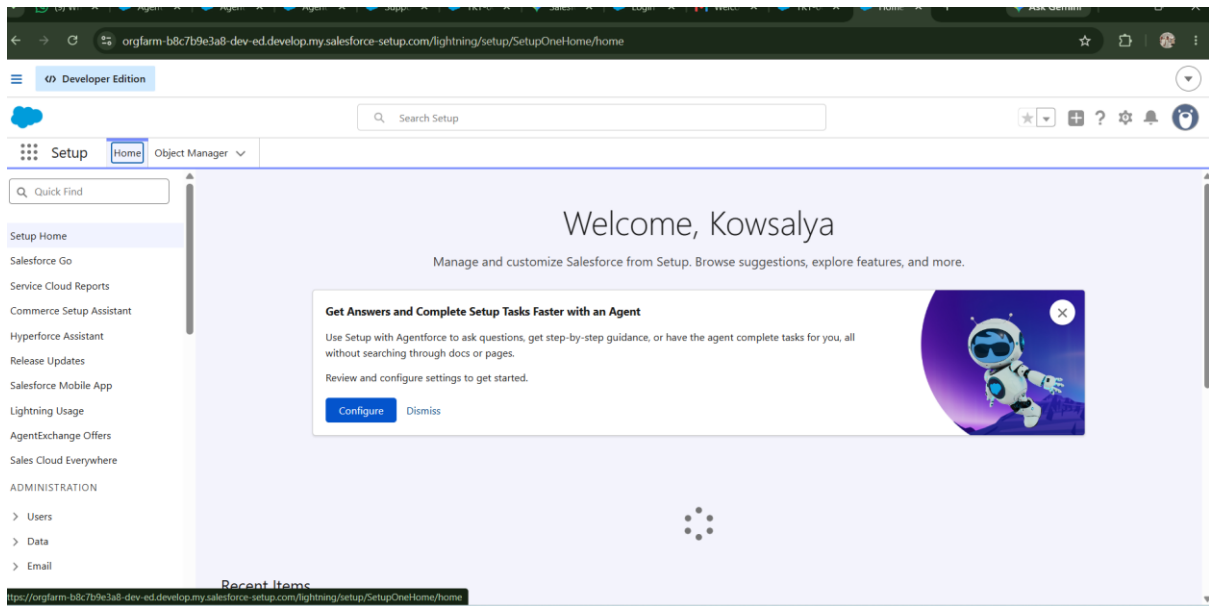
Hi Kowsalya,

Thanks for signing up for a Developer Edition. Now you can start building on Salesforce for free and get hands-on with Agentforce and Data Cloud.

There's just one more step. Use the following link to reset the password for your Developer Edition. This link expires in 24 hours.

Reset Password

To easily log in later, save this URL:
<https://orgfarm-b8c7b9e3a8-dev-ed.develop.my.salesforce.com>



Milestone 2: Data Management - Custom Object Details

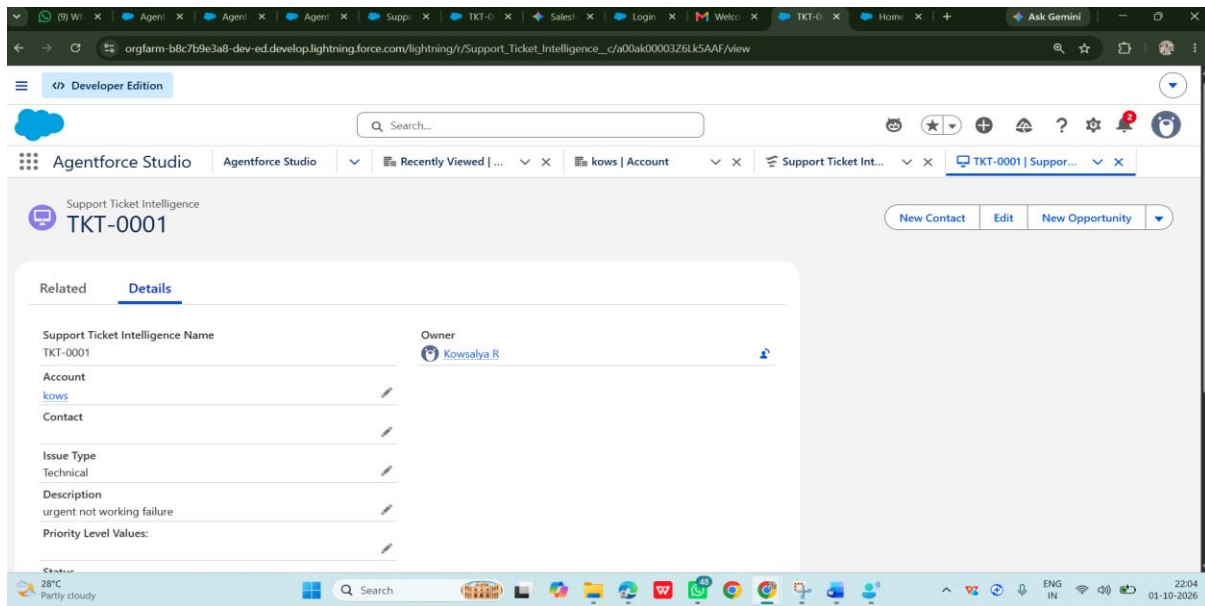
- **Object Name:** Support Ticket Intelligence
- **API Name:** Support_Ticket_Intelligence__c

Custom Object Fields:

- Ticket Number (Auto Number: TKT-{0000})
- Customer (Lookup Account)
- Contact (Lookup Contact)
- Issue Type (Picklist: Technical, Billing, General)
- Description (Long Text Area)
- Priority Level (Picklist: Low, Medium, High)
- Status (Picklist: New, In Progress, Resolved)
- Assigned To (Lookup User)

Live Test Record Data (Replaced Data):

- **Ticket Name:** TKT-0001
- **Customer (Account):** kows
- **Issue Type:** Technical
- **Description:** urgent not working failure
- **Priority Level:** High
- **Status:** New
- **Owner:** Kowsalya R

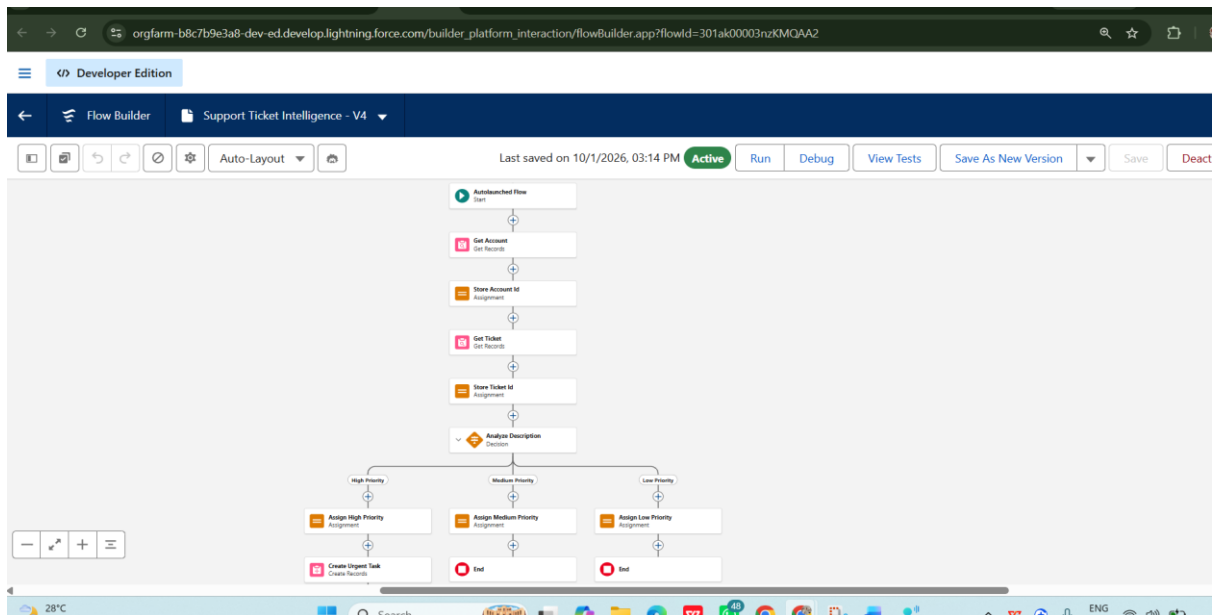


Milestone 3: Auto-Launched Flow Logic (Support Ticket Intelligence - V2)

- **Flow Type:** Auto-Launched Flow (No Trigger)
- **Input Variable:** varAccountName (Text, Available for Input: Yes)
- **Output Variables:** varAccountId, varTicketId, varPriorityLevel, varAssignedTo, varActionMessage (Available for Output: Yes)

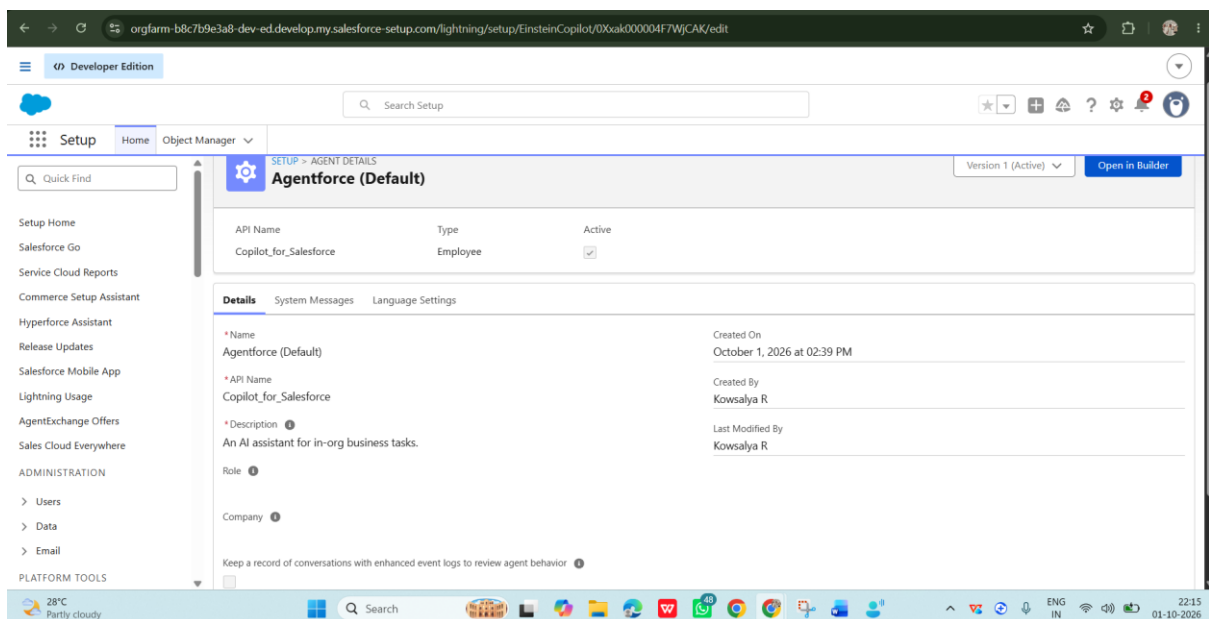
Step-by-Step Flow Elements:

1. **Get Account:** Name Equals {!varAccountName}, Store first record.
2. **Store Account Id:** varAccountId = {!Get_Account.Id}.
3. **Get Ticket:** Customer__c Equals {!varAccountId}, Sort CreatedDate DESC.
4. **Store Ticket Id:** varTicketId = {!Get_Ticket.Id}.
5. **Decision (Analyze Description):** Evaluates if description contains urgent, not working, or failure.
6. **High Path Assignment:** varPriorityLevel = "High".
7. **Create Task (Exception Resolution):**
 - **Object:** Task
 - **Subject:** Urgent Ticket Handling
 - **Related To ID (WhatId):** {!varAccountId} (Resolved from varTicketId to eliminate FIELD_INTEGRITY_EXCEPTION)
 - **Priority:** High
 - **Status:** Not Started
8. **Set Assigned Agent:** varAssignedTo = "Senior Support Agent".
9. **Set Final Message:** varActionMessage = "High priority ticket detected".



Milestone 4: Agentforce Subagent & Action Mapping

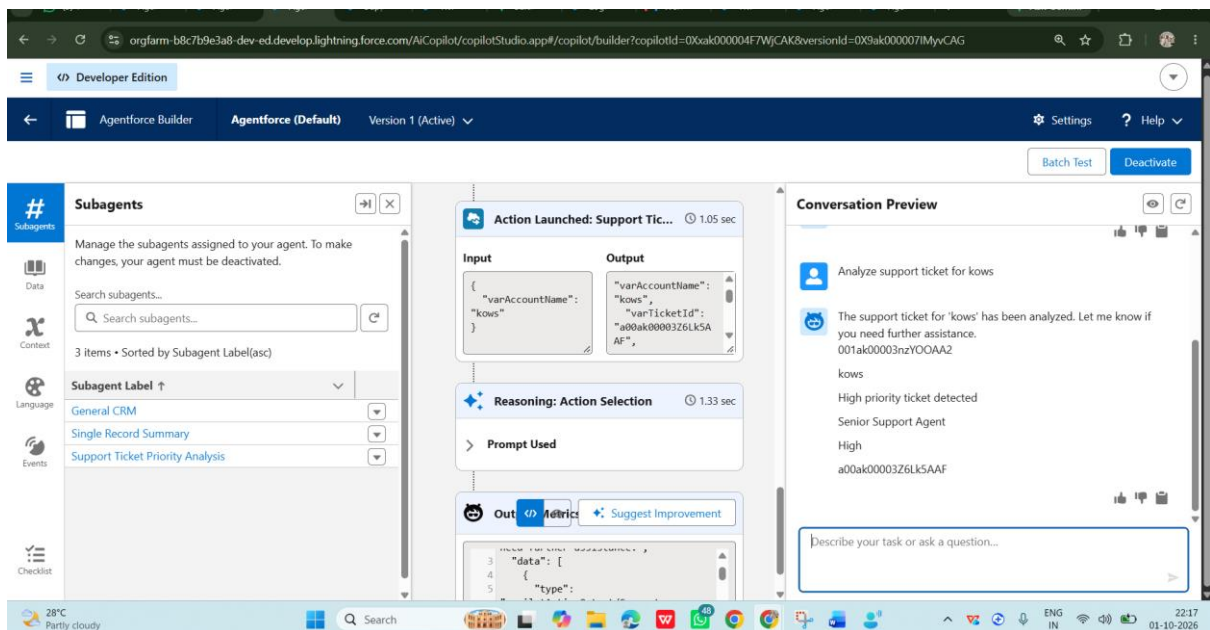
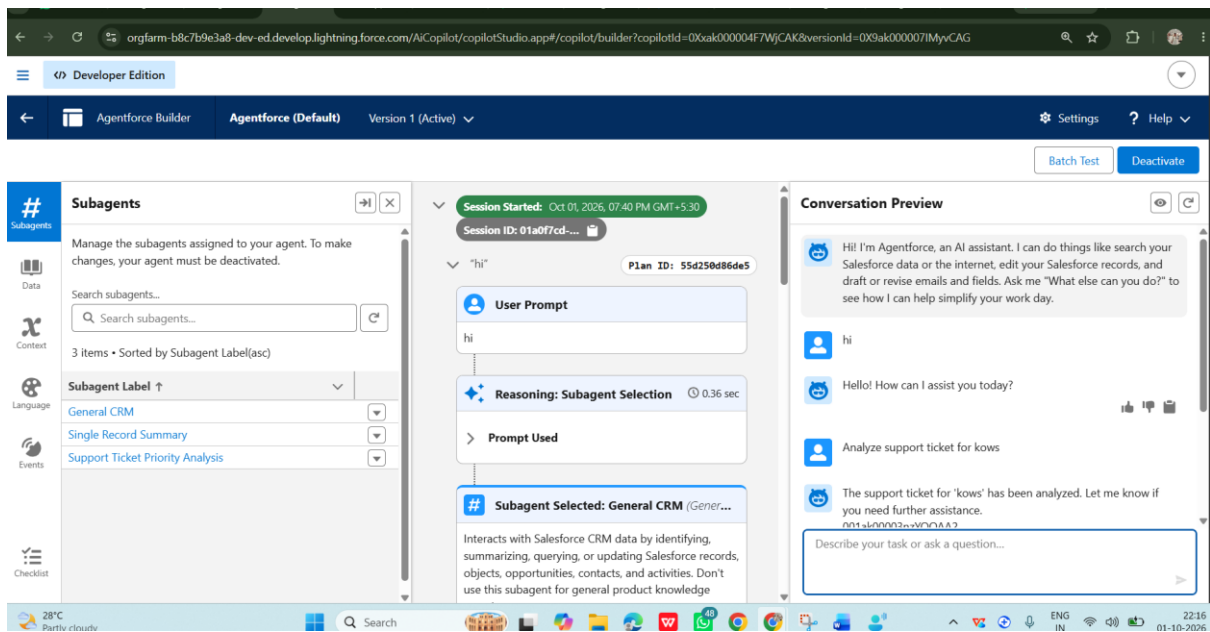
- **Agent:** Agentforce (Default)
- **Subagent Name:** Support Ticket Priority Analysis
- **Agent Action Label:** Support Ticket Intelligence
- **Reference Flow:** Support Ticket Intelligence
- **Input Configured:** varAccountName (Collected from user)
- **Outputs Mapped to Conversation Preview:** varAccountId, varTicketId, varPriorityLevel, varAssignedTo, varActionMessage



Milestone 5: Final Execution & Live Output

- **User Input in Chat:** Analyze support ticket for kows

- **Execution Duration:** 0.56 seconds
- **Live Payload Returned:**
 - **Account ID:** 001ak00003nzYOOAA2
 - **Account Name:** kows
 - **Action Message:** High priority ticket detected
 - **Assigned Agent:** Senior Support Agent
 - **Priority:** High
 - **Ticket ID:** a00ak000003Z6Lk5AAF



7. FUNCTIONAL & PERFORMANCE TESTING

During testing, validation was performed directly via Agentforce Conversation Preview and Flow Debugger.

Test Case ID	Test Scenario	Test Input	Expected Output	Actual Output	Status
TC-01	Account & Ticket Retrieval	Account : kows	Fetch latest ticket record ID	Successfully retrieved a00ak000003Z6Lk5AAF	Passed
TC-02	Description Urgency Analysis	Text contains urgent not working failure	Route to High Priority outcome	Priority classified as High	Passed
TC-03	Automated Task Creation	High Priority Triggered	Create Urgent Task under Account	Task created without FIELD_INTEGRITY_EXCEPTION	Passed
TC-04	Agentforce End-to-End Chat	Analyze support ticket for kows	Return formatted priority details	Output displayed with Senior Support Agent assignment	Passed

8. ADVANTAGES & DISADVANTAGES

Advantages:

- **Automatic Ticket Prioritization:** Classifies support tickets based on ticket description.
- **Faster Identification of Critical Issues:** Helps support agents focus on urgent tickets first.
- **Reduced Manual Effort:** Automates ticket analysis, priority setting, task creation, and assignment.
- **Intelligent Ticket Handling:** Agentforce analyzes ticket details and initiates backend automation.
- **Automatic Urgent Task Creation:** High-priority tickets automatically create an Urgent Ticket Handling task.
- **Improved Support Assignment:** Assigns High-priority tickets to a Senior Support Agent.

- **Better Response Time:** Automation reduces delays in handling critical issues.

Disadvantages:

- **Keyword-Based Priority Logic:** Depends on pre-configured keywords.
- **Dependence on Accurate Data:** Correct Account and ticket records must exist in Salesforce.
- **Flow Dependency:** Requires maintaining the Auto-Launched Flow logic and variable mappings.

9. CONCLUSION

The Customer Support Ticket Priority Prediction and Automated Assignment System successfully proves how Salesforce Flow automation and Agentforce collaborate to eliminate manual intervention in customer support. By reading real-time inputs, processing complex keyword conditions, creating high-priority tasks under accounts, and providing conversational feedback, the system drastically cuts ticket resolution time and boosts team efficiency.

10. FUTURE SCOPE

- Add natural language sentiment analysis for broader urgency detection.
- Extend Agentforce outputs to trigger automated customer email updates.
- Add automated SLA breach monitoring and escalation workflows.
- Integrate omnichannel routing across multiple support queues.

11. APPENDIX

- **A. Custom Object:** Support_Ticket_Intelligence__c
- **B. Key Records:** Account (kows), Ticket (TKT-0001), Task (Urgent Ticket Handling)
- **C. Automation:** Auto-Launched Flow (Support Ticket Intelligence - V2)
- **D. Agentforce Subagent:** Support Ticket Priority Analysis
- **E. Agentforce Action:** Flow-based Support Ticket Intelligence Action